seeed studio

INDUSTRIAL PRODUCT DATASHEET

SO-ARM101 Pro Unassembled Bundle – LeRobot Compatible 6DOF Robotic Arm

SO-ARM101 Pro Unassembled Bundle: an open-source, LeRobot-compatible 6DOF robotic arm for imitation learning, teleoperation, and cloud-powered robot AI training.

SKU	UPDATED	PRICE
E26070901	2026-10-08 09:47:30	$0.00(Excl. VAT)



Overview



The SO-ARM101 Pro Unassembled Bundle is a LeRobot-compatible 6DOF robotic arm featuring six STS3215 magnetic-encoder bus servos with 30 kg·cm torque per joint and 12-bit absolute position feedback for smooth, quiet, and repeatable motion. It offers affordable 3D-printable kit options, fully open-source hardware and software, and guided setup through the SenseCraft Robotics platform for data collection and cloud-powered robot AI training.

Key Features

【Note】

- From China warehouse to the United States, choose "**US Direct Group Shipping**" to enjoy duty-free！
- **6 Degree-of- Freedom(DOF) Arm** — Six STS3215 magnetic-encoder bus servos deliver 30 kg·cm torque per joint with 12-bit absolute position feedback for smooth, quiet, and repeatable motion
- **Hugging Face LeRobot Native Support** — Works out of the box with the LeRobot framework for imitation learning; collect teleoperation data and train ACT/Diffusion models with community-shared pipelines.
- **Affordable & Flexible Kit Options** — Low-cost 3D-printable robotic arm with flexible kit options. Select full-assembly, servo-only pack or unassembled parts to match your budget and project needs.
- **Fully Open-Source** — All hardware designs, BOM, firmware, and software are publicly available; modify, extend, and share freely

One-Stop Guided Setup & Cloud-Powered Robot AI Training

SenseCraft Robotics is an in-house developed, exclusive platform from Seeed Studio. It delivers an all-in-one robot-AI workflow featuring guided setup, cloud training and lightweight local validation. It lets you finish essential pre-training steps — device pairing, serial binding, camera binding, calibration check and teleoperation check — all in one place, greatly streamlining your setup for education labs, R&D PoCs and early-stage robot validation projects.
Platform download ：
https://sensecraft.seeed.cc/en/robotics?utm_source=bazaar&utm_medium=ecommerce&utm_campaign=robotics_hardware&utm_content=product_detail

Product Gallery







Specification

Arm Type	SO-ARM101	
	Leader Arm	Follower Arm
Motor Configuration	1 × 7.4 V motor, 1:345 gear ratio — Joint 2	12x (12V) 1:345 gear ratio motors for all joints
	2 × 7.4 V motors, 1:191 gear ratio — Joints 1 & 3	
	3 × 7.4 V motors, 1:147 gear ratio — Joints 4 & 5 & Gripper	
Power Supply	DC 5V3A	DC 12V2A
DoF	5+1	
Payload	500g	
Angle sensor	12-bit magnetic encoder	
Operating Temperature Range	0℃～40℃	
Communication Method	UART	
Control Method	PC	

	SO-ARM101 Pro Servo Motor Kit	SO-ARM101 Pro Unassembled Bundle	SO-ARM101 Pro Kit
Price	$189.00	$295.00	$299.00
Assembly	Structural parts to be 3D-printed by user	Assembly from cratch	Pre-assembled
USB Camera	Excluded	Excluded	Included
Best For	Users with existing 3D-printed frames or custom enclosures	Hobbyists who enjoy the build process and want a complete kit. Select accessories as you like.	Researchers, educators, and teams who want to skip the build and focus on AI
Time to First Experiment	8+ hours	2-3 hours	Under 30 minutes

Hardware Overview





Applications

- **Leader-Follower Teleoperation:** Collect robot demonstration data easily via leader-follower teleoperation, ready for imitation-learning model training.
- **Imitation Learning & Reinforcement Learning:** Teach your robot household-style skills like folding clothes with human demos. Gather data via leader-follower teleoperation, train imitation-learning policies on LeRobot, and watch your robot reproduce tricky real-world manipulation tasks. SO-101 works seamlessly with LeRobot for data collection, simulation, training, and deployment. Train and deploy custom grasp-and-place policies quickly on NVIDIA Jetson edge devices. Explore the tutorial: LeRobot on NVIDIA Jetson AI Lab
- **Community Inspiration:** Follow Nikodem Bartnik's hands-on SO-101 workflow: hardware assembly, servo control, custom policy training and real-time block-picking inference. Walk through key steps, fix typical issues and explore the open-source arm's potential. Our monthly project call is now open! Share your project demos for a chance to win prizes and discount coupons.
- **Quick Start Developer Resources:** Jumpstart your project with official tutorials, open-source repositories, and community support: Start building your first Physical AI Project in 30 mins with SenseCraft; Step-by-Step Guide: SO-ARM 101 Quick Start; Open-Source Code: Access PyTorch models, simulation tools, and datasets on the Hugging Face LeRobot GitHub Repository; Community & Support: Join discussions, share models, and get real-time help on the Seeed Discord Channel & FaceBook. Join our Facebook Group to claim exclusive coupons worth up to $100 USD.

Part List

7.4v STS3215 Servo Motor 1:345 gear rate	x1
7.4v STS3215 Servo Motor 1:191 gear rate	x2
7.4v STS3215 Servo Motor 1:147 gear rate	x3
12v STS3215 Servo Motor 1:345 gear rate	x6
Servo Driver Board for LeRobot	x2
Stud	x8
Screw	x8
5V Power Supply Cable (Multiple head)	x1
12V Power Supply Cable (Multiple head)	x1
USB-C Cable	x2
DC Power Pigtail Cable	x2
Fixing clip	x4

Leader Parts	12 pcs
Follower Parts	11 pcs
Clamps Parts	12 pcs

32*32mm USB Camera	x1

Resource Links

- https://www.seeedstudio.com/SO-ARM101-Pro-Unassembled-Bundle.html
- STS3215 Servo Motor Datasheet
- Finedtune 3D printed parts files
- Original SO-ARM101 3D printed parts
- Servo Adaptor Board for Seeed Studio XIAO Datasheet
- reComputer J301x Datasheet
- reComputer J401X Datasheet
- reComputer_mini_datasheet_V1.0

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